

Summary Report of a Public Workshop: Case Studies of Multi-Regional Clinical Trial Incorporating Concept of the ICH E17 Guideline

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To further our understanding of the International Council for Harmonization of Technical Requirements for Pharmaceuticals for Human Use (ICH) E17 guideline and promote effective implementation, a public workshop was held in Japan by regulatory agency and industry representatives. In this workshop, important concepts explained in the ICH E17 guideline, such as intrinsic/extrinsic ethnic factors that influence treatment effects (“effect modifiers”) and the holistic evaluation of consistency of treatment effect were actively discussed through case studies. The importance of holistic evaluation of consistency was recognized, and it was concluded that the evaluation and relevant discussion should be shared with regulatory authorities, sponsors, and broader stakeholders.

BACKGROUND AND PURPOSE OF WORKSHOP

Recently, many multi-regional clinical trials (MRCTs) have been conducted. The MRCT collects data from multiple regions simultaneously; therefore, it has the advantage of evaluating the benefits and risks of drugs in different regions and ethnic groups, as well as increasing efficiency of drug development. Variability of intrinsic/extrinsic factors may cause differences in the benefits and risks of drugs between regions or ethnic groups. In the International Council for Harmonization of Technical Requirements for Pharmaceuticals for Human Use (ICH) E17 guideline, which reached Step 4 in 2017, to plan a structured exploration to examine the consistency of treatment effects across regions and subpopulations is recommended. The guideline emphasizes that consistency needs to be evaluated in a holistic manner, stating that “biological plausibility,” “internal consistency,” “external consistency,” “clinical relevance,” and “statistical uncertainty” (they are defined as “five viewpoints for consistency” in this report) should be considered to support the overall findings.¹ Thus, the MRCT working group in Japan, which comprises industrial members [Japan Pharmaceutical Manufacturers Association (JPMA), Japan-based committee of Pharmaceutical Research and Manufacturers of America, and European Federation of Pharmaceutical Industries and Associations in Japan] and regulatory agency members [Pharmaceuticals and Medical Devices Agency (PMDA)] held the public workshop in 2022 with the aim of deepening the

understanding of holistic and structured evaluation of consistency based on the concept of the ICH E17 guideline using several case studies.

CASE STUDIES BY THE PMDA

Vericiguat tablets² and finerenone tablets,³ which have been approved based on confirmatory MRCTs in Japan in 2021 and 2022, respectively, were discussed as case studies to deepen understanding of strategies for consistency evaluation in MRCTs. The results of the primary end point in the Japanese subpopulation and the overall population were compared to evaluate consistency. The primary end points in the MRCTs of vericiguat (study 16493) and finerenone (studies 16244 and 17530) were a composite of cardiovascular events (studies 16493 and 17530) or renal events (study 16244), and these MRCTs were large-scale studies. In these studies, consistent results were obtained in the overall population and the Japanese population for the primary end point (composite end point); however, for the clinically important components, cardiovascular death (vericiguat) or kidney failure (finerenone), inconsistent results were observed between the overall and Japanese populations. In addition to considering statistical uncertainty due to the small and limited Japanese sample size, a holistic evaluation of consistency considering biological plausibility, internal consistency, and external consistency was conducted to support the judgment that efficacy can be expected in the regional population.^{2,3}

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These cases show that consistency should be evaluated multidimensionally and systematically, and should not be concluded based only on a single assessment.

CASE STUDIES BY THE INDUSTRY

Important concepts explained in the ICH E17 guideline and its training material, such as intrinsic/extrinsic ethnic factors that may influence treatment effects (referred to as “effect modifiers”) and the holistic evaluation of consistency, were introduced through case studies. To deepen the understanding of the concepts, a “3-layer approach” which was previously proposed by a task force of the Data Science Expert Committee of the JPMA,⁴ was explained. In this approach, the results of MRCTs are evaluated first in the overall population (layer 1), and then the results of subgroup analyses are evaluated to determine whether there are inconsistencies between a specific subgroup and the overall results, and whether there are the factors that may affect the results, that is, effect modifiers (layer 2). The benefit–risk balance for a country should be discussed based on the effect modifiers identified in layer 2 (layer 3). This concept attempts to explain the regional differences observed in MRCTs by exploring effect modifiers using overall data, and to estimate treatment effects for each region, which have a high affinity with the ICH E17 guideline. That is, a primary analysis for estimation of the overall treatment effect should be planned, the information on intrinsic and extrinsic factors important to interpret the treatment effect should be collected and their impact should be examined, and the evaluation of the consistency of treatment effects across regions and subpopulations should be also included in the analyses.

Secukinumab injection, pertuzumab injection, palbociclib capsules, and blonanserin transdermal patches were discussed as case studies. These drugs were approved in Japan between 2013 and 2020, based on confirmatory MRCTs. Japanese participants had been enrolled in these studies, and the consistency of treatment effects between the overall population and Japanese subpopulations had been evaluated (Table 1). In the case studies at the workshop, retrospective assessments based on the ICH E17 guideline were provided: The case studies focused on layer 2, aiming to identify effect modifiers in efficacy and assess their impact on the efficacy results. The results in the Japanese subpopulation were then interpreted, evaluating the benefit–risk balance based on an understanding of the effect modifiers or their absence (layer 3). Subgroup analyses were conducted to explore the effect modifiers, and additional analyses were conducted when feasible. The holistic evaluation of the consistency was also discussed in the workshop (Table 2).

Case 1: Secukinumab injections (PREVENT⁵)

The clinical relevance and results of subgroup analyses of the PREVENT study⁶ were evaluated to identify potential effect modifiers. Objective signs of inflammation, as evidenced by elevated C-reactive protein (CRP) levels or magnetic resonance imaging (MRI) findings, were the most important prognostic indicators of non-radiographic axial spondyloarthritis stratified by randomization. Subgroup analysis showed higher treatment differences in CRP+ vs. CRP– and MRI+ vs. MRI– patients (Figure S1), particularly the double-positive subgroup (CRP+/MRI+) vs.

the single-positive groups (CRP+/MRI– or CRP–/MRI+). Male participants were more prone to disease progression; however, the subgroup analysis showed that male participants had better responses than female participants. Other effect modifiers included human leukocyte antigen-B27 status, body weight, previous experience with tumor necrosis factor inhibitor treatment, and baseline severity measured using the Bath Ankylosing Spondylitis Disease Activity Index.

Post hoc analyses of pooling regions or subpopulations were exploratory attempted. Due to the small number of patients in each participating country, countries with a similar distribution of effect modifiers were clustered into five pooled regions (Data S1, Table S1). Positive treatment differences were observed in each region except for region 2, which may have been due to the small sample size in the region. The characteristics of the participants in region 1, including Japan, showed a lower proportion of CRP+ patients and a higher proportion of MRI+ patients. The treatment difference in region 1 was smaller than that in the overall population. The treatment difference was largest in region 5, which had the highest proportion of CRP+/MRI+ participants, consistent with the findings in the subpopulation above. In contrast, region 4 showed a large treatment difference despite having the lowest proportions of CRP+/MRI+ and male patients. Low response rates across treatment groups were observed, and all countries (France, Germany, Italy, and the Netherlands) in region 4 are geographically close, suggesting the presence of factors specific to this region. Participants were also classified into 10 subpopulations and differences in treatment effects by subgroups of CRP/MRI status and sex combinations were determined (Table S2).

The assessment clarified the effect modifiers and their influence on regional differences. The treatment differences were generally consistent among the pooled regions for commonalities in the effect modifiers, but differed slightly depending on the distribution of CRP+/MRI+ and sex (Figure S2). The inconsistent results observed in the Japanese subpopulation could be due to an imbalance in treatment assignment and a small sample size (2 participants in the secukinumab arm and 7 in the placebo arm), but not due to potential effect modifiers. These assessments indicated that the treatment effects observed in the overall population could be applied across any region, including Japan.

Case 2: Pertuzumab injections (CLEOPATRA⁷)

Because no apparent effect modifiers were identified in the human epidermal growth factor receptor 2-enriched population, the analysis focused on the comprehensive subgroups defined using intrinsic/extrinsic factors to evaluate consistency and identify potential effect modifiers. Prespecified subgroup analyses for progression-free survival (PFS) and overall survival showed consistent benefits of pertuzumab in multiple subgroups (Figures S3, S4). This reproducibility across multiple end points supported internal consistency, suggesting the robustness of the benefits across multiple patient attributes. Although participants with non-visceral diseases showed a hazard ratio (HR of approximately 1.0), this should be interpreted with caution considering the unclear clinical relevance and statistical uncertainty. These findings were consistent with the fact that no clear biological evidence suggested a

Table 1 Summary of studies for case studies by industry

Case drug	Case 1. secukinumab injection	Case 2. pertuzumab injection	Case 3. palbociclib capsule	Case 4. blonanserin transdermal patch
Year of approval in Japan	2020	2013	2020	2019
Confirmatory study	PREVENT ⁵	CLEOPATRA ⁷	PALOMA-3 ⁹	D4904020 ¹⁰
Study design and objectives	A randomized, double-blind, placebo-controlled study that evaluated the efficacy, safety, and tolerability of secukinumab, with LD or NL, in patients with active nr-axSpA ⁵	A randomized double-blinded placebo-controlled trial comparing pertuzumab, trastuzumab, and docetaxel combination therapy (pertuzumab group) with trastuzumab and docetaxel combination therapy (control group) in patients with HER2-positive MBC who had not received previous chemotherapy or biological treatment for their metastatic disease ⁷	A randomized, double-blind, phase III study that evaluated the efficacy and safety of the combination of palbociclib and fulvestrant compared with fulvestrant alone in premenopausal or postmenopausal women with HR-positive, HER2-negative (HR+/HER2-) MBC whose disease had progressed on prior endocrine therapy ⁹	A randomized, double-blind, multicenter, placebo-controlled, phase III study to determine the efficacy, safety, and pharmacokinetics of a blonanserin transdermal patch (Lonasen Patch) in participants with acute exacerbation of schizophrenia ¹⁰
Study end point	The primary end point at week 16 was a ASAS40 criteria ¹⁵ in tumor necrosis factor inhibitor-naïve patients, with that in the overall population as a secondary end point	The primary end point of this study was the IRF-PFS, and OS was a key secondary end point	The primary end point was investigator-assessed PFS, according to RECIST version 1.1	The primary end point was the change from baseline in PANSS total score after 6 weeks of treatment
Number of participants	A total of 555 eligible participants from 24 countries were enrolled, including 13 Japanese participants	A total of 808 participants were enrolled from 25 countries, including 53 participants from Japanese centers ⁸	A total of 521 participants from 17 countries were enrolled, including 35 participants from Japanese centers ^{9,16}	A total of 580 participants from 7 countries, including 164 Japanese participants, were randomized
Key efficacy results	The ASAS40 response rates in the overall population were significantly higher in the LD (40.0%) and NL (40.8%) groups than in the placebo group (28.0%). In the Japanese subpopulation, the ASAS40 response rates at week 16 for the LD, NL, and placebo groups were 0% (0/2), 50% (2/4), and 14.3% (1/7), respectively. Although the ASAS40 response was not observed in the LD group, the ASAS40 response in the pooled secukinumab group was 33.3% (2/6), which was higher than 14.3% (1/7) in the placebo group.	The superiority of pertuzumab compared with the control in IRF-PFS was demonstrated with a stratified hazard ratio of 0.62 (95% CI: 0.51–0.75). Following the primary analysis, the superiority of OS has also been demonstrated in the second interim analysis for OS. ¹⁷ Prespecified subgroup analyses of PFS/OS also showed consistent benefits. In the Japanese subpopulation, however, IRF-PFS in the Japanese subpopulation showed an inconsistent trend, with a hazard ratio of 1.63 (95% CI: 0.70–3.78). ⁸	The study was stopped early for efficacy based on the results of a preplanned IA after 195 events had occurred. The study met its primary end point at the IA (hazard ratio of 0.42, 95% CI: 0.32–0.56, $P < 0.001$). ⁹ In the Japanese subpopulation, 6 PFS events occurred (6 events in the palbociclib group and no event in the placebo group) at the time of IA, and the hazard ratio of the PFS was not lower than 1.	The differences in the LSM change in the PANSS total score from baseline compared with placebo was –5.6 for blonanserin 40mg (95% CI: –9.6 to –1.6; adjusted $P = 0.007$) and –10.4 for blonanserin 80mg (95% CI: –14.4 to –6.4; adjusted $P < 0.001$) based on the mixed model for repeated measurements. The observed difference between blonanserin 40mg and placebo in Japan was smaller than the overall result.

Abbreviations: ASAS40, 40% improvement in disease activity according to the Assessment of Spondylo-Arthritis International Society; CI, confidence interval; HER2, human epidermal growth factor receptor 2; HR, hormone receptor; IA, interim analysis; IRF, independent review facility; LD, loading dose; LSM, least squares mean; MBC, metastatic breast cancer; NL, no loading dose; nr-axSpA, non-radiographic axial spondyloarthritis; OS, overall survival; PANSS, positive and negative syndrome scale; PFS, progression-free survival; RECIST, Response Evaluation Criteria in Solid Tumors.

difference among races in the efficacy and safety of pertuzumab, as reported to date. Although the HR was > 1 in the Japanese subpopulation,⁸ the limited sample size and imbalance in the important prognostic factors between the treatment groups may have caused such a result.

Case 3: Palbociclib capsules (PALOMA-3⁹)

The study met its primary end point (PFS) at the preplanned interim analysis (IA; HR: 0.42 with 95% confidence interval [CI]: 0.32–0.56, $P < 0.001$).⁹ Consistent results were observed across subgroups according to stratification, demographics, and

Table 2 Assessment of the consistency of treatment effects in each case

Case drug	Case 1. secukinumab injection	Case 2. pertuzumab injection	Case 3. palbociclib capsule	Case 4. blonanserin transdermal patch
Biological plausibility	No major differences in disease characteristics and diagnostic criteria among regions. No apparent difference in PK profile among races. Minimized number of patients previously treated with TNFi, which could affect treatment effects.	No reported differences in the efficacy and safety of pertuzumab in specific populations, except for HER2 status. No apparent difference in PK profile among ethnic groups.	No major differences in disease characteristics among regions. No intrinsic factor caused a major difference in the PK profile.	No clinical or pharmacological features led to regional differences
Internal consistency	Generally consistent treatment effect across the pooled regions for commonalities of effect modifiers. Slight difference depending on the distribution of CRP/MRI abnormalities and sex.	Consistent treatment effect regardless of subgroups. Consistency suggested across multiple efficacy end points.	Consistent PFS results regardless of subgroups	Consistent results suggested in multiple efficacy end points. Efficacy suggested regardless of subgroups.
External consistency	Findings on the effect of CRP/MRI abnormalities and sex on treatment effect are consistent with studies of TNFi ⁶	<i>Trastuzumab with a similar target has shown consistent efficacy and safety regardless of intrinsic or extrinsic ethnic factors</i>	NA (at the time of the IA)	Similar results with oral formulation
Clinical relevance	CRP/MRI abnormalities and sex were considered clinically relevant factors	Little evidence to suggest that pertuzumab is less effective in the non-visceral disease subgroup	Hazard ratio < 1 in all subgroups with no clinically meaningful differences	No clinically relevant difference among countries/regions
Statistical uncertainty	Relatively small number of patients in some pooled regions and subgroups. The results in the Japanese population are considered by chance due to the imbalanced treatment assignment and small sample size rather than an effect of CRP/MRI abnormalities and sex.	No clear difference among subgroups considering the variability	Hazard ratio < 1 in all regions and races. Small number of subjects included and large variability in some subgroups.	A small number of subjects in some countries/regions. Consistency among countries/regions cannot be ruled out in multiple analyses.

Abbreviations: CRP, C-reactive protein; HER2, Human epidermal growth factor receptor 2; IA, interim analysis; MRI, magnetic resonance imaging; NA, not applicable; PFS, progression-free survival; PK, pharmacokinetics; TNFi, tumor necrosis factor inhibitor. Italicized text indicates information known prior to the conduct of the study.

prognostic factors. In the Japanese subpopulation, six PFS events (6 events in the palbociclib group and no events in the placebo group) occurred at the time of IA, and the HR of PFS was not < 1. In contrast, the efficacy in Asian participants ($N=105$) was consistent with that in the overall population (HR in Asians, 0.70, 95% CI: 0.34–1.45). In the United States and the European Union, new drug application (NDA) submissions for palbociclib was completed based on the IA results of the PALOMA-3 study. Double blinding was maintained after IA, and two additional analyses were conducted. The NDA in Japan was submitted based on the results of the second additional

analysis in the Japanese subpopulation, which showed a consistent trend with the overall population (HR in Japanese, 0.82, 95% CI: 0.32–2.11).

To evaluate the overall consistency of efficacy, analyses were performed to evaluate the treatment effects across subgroups according to stratification, demographics, and prognostic factors (Figure S5). Subgroup analyses showed that the effects were consistent with the overall results. Preplanned subgroup analyses by geographic region (North America, European Union, and Asia/Pacific) and race (White, Asian, Black, and others) also revealed consistent results. All subgroup analyses suggested that there were

no effect modifiers that caused regional differences in the treatment effects. Thus, it was concluded that the clinical relevance of palbociclib in the Japanese subpopulation was consistent with that in other populations at the time of IA.

Case 4: Blonanserlin transdermal patches (D4904020¹⁰)

The dose-dependent differences in the least squares mean change in the Positive and Negative Syndrome Scale (PANSS) total score from baseline was observed based on the mixed model for repeated measurements (Table 1). The observed difference between blonanserlin 40 mg and the placebo in Japan was smaller than the overall result, whereas the positive dose-dependency was not observed in China, Korea, and Ukraine (Figure S6).

A subgroup analysis based on the demographics and psychiatric history of the participants was conducted and reviewed to identify effect modifiers and explain regional variations. Although the duration of illness could be expected to be an effect modifier, it was concluded not to be an effect modifier after a thorough review, including scatter plots with changes in the PANSS total score. Eventually, it was impossible to explain regional inconsistencies using any effect modifier.

Additional analyses, such as tests for efficacy and regional interaction,¹¹ and reviews using the James–Stein shrinkage estimator¹² were conducted. These additional analyses confirmed no efficacy changes or regional interactions; therefore, inconsistencies in regional efficacy, including Japan, were suggested (Figure S7).

DISCUSSION

As the number of MRCTs increases, more applications for drug approval are being submitted based on the MRCT results. However, because bridging strategies in accordance with the ICH E5 guideline¹³ were mainly adopted before MRCTs were widely conducted, data from each country/region in an MRCT have often been discussed in terms of the consistency of treatment effects between the population from a specific country/region and the overall population. For certain countries/regions, interest may be directed toward how data from that country/region behave with respect to efficacy and safety. In today's MRCTs, as a result of the increase in participating countries/regions worldwide, the number or percentage of participants from each country/region would be decreasing. Under these circumstances, it is more realistic and scientifically sound to conduct holistic evaluations of consistency. In this workshop, both the PMDA and industries holistically assessed the consistency in each case study.

Even prior to the implementation of the ICH E17 guideline, “subgroup analyses” and “the evaluation of consistency in treatment effects across regions, race, or variant frequencies” have been commonly conducted in the evaluation of MRCTs.¹⁴ In the case studies from industry, maximum use of the analyses and interpretation of the results with an understanding of “effect modifiers” was proposed.

Notably, the conclusions regarding benefit–risk in regions did not change with the *post hoc* analyses; however, the approach provided more insights. An explanation using effect modifiers can be useful when reviewing all the information obtained in a drug development program. The benefit–risk could be more precisely

evaluated if effect modifiers are identified and if the distribution of effect modifiers in the local patient population is known. Therefore, the 3-layer approach, which first evaluates the results of the overall population, then searches for effect modifiers based on the results of subgroup analyses, and last evaluates benefit–risk balance in each country, is useful for consistency evaluation and discussing the benefit–risk balance in Japan and in other countries or regions.

The holistic evaluation of consistency could be further improved with properly planned analysis, considering predefined effect modifiers and findings from multiple data sources, such as clinical trials for other study populations. As these findings accumulate over time throughout the clinical development program, it is also important to continue to investigate effect modifiers and their potential impact as development proceeds. When the factor is used as one of the stratification factors for randomization, the subgroup analysis based on the factors have greater confidence, and when the factor is not defined at the design stage, interpretation of the results may be difficult.

When evaluating MRCT data for regulatory submissions, sponsors should elaborate on the process of evaluating and interpreting consistency of treatment effect. Common Technical Documents (CTDs) of NDAs should include a holistic evaluation of consistency, the results of an investigation of effect modifiers, and details of the analyses conducted for consistency evaluation. A more systematic explanation in accordance with ICH E17 guideline may be needed in review reports to facilitate the utilization of MRCT data. Improvement in structure and/or contents of CTDs and review reports from the perspective above would lead to efficient communication among sponsors and regulatory authorities, resulting in the appropriate interpretation of MRCT results for each region.

CONCLUSIONS

For the efficient and successful development of a new drug, the planning and implementation of MRCTs in accordance with ICH E17 guideline and appropriate evaluation of MRCTs results are crucial. This workshop aimed to evaluate the MRCTs results based on the concept of ICH E17 guideline. Industry's case studies provided detailed considerations on how to explore the potential effect modifiers and how to evaluate the efficacy of a drug in a patient population in a country/region based on the findings related to effect modifiers. A holistic evaluation of consistency is important to discuss the benefit–risk balance, and regulators should be informed on what is clear or unclear from five viewpoints for consistency and their corresponding limitations.

The holistic evaluation of consistency and relevant discussion should be clearly described in the CTDs and the review reports by the sponsor and PMDA, respectively. It enables regulators, sponsors, and broader stakeholders, such as academia and healthcare practitioners, to facilitate the transparent and comprehensive interpretation of the MRCT results.

SUPPORTING INFORMATION

Supplementary information accompanies this paper on the *Clinical Pharmacology & Therapeutics* website (www.cpt-journal.com).

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CONFLICT OF INTEREST

All the authors declared no competing interests for this work. All authors are employees of their organizations listed in the author affiliations.

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